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STANDARDS AND GOVERNANCE

MSS-GOV

USAREUR-AF MSS NAMING AND GOVERNANCE STANDARDS

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HEADQUARTERS
UNITED STATES ARMY EUROPE AND AFRICA
(USAREUR-AF)
Wiesbaden, Germany

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USAREUR-AF MSS NAMING AND GOVERNANCE STANDARDS

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Proponent: USAREUR-AF C2DAO (Command and Control Data and Analytics Office) **Authority:** Army CIO Data and Analytics Policy (April 2024); UDRA v1.1 (February 2025) **Distribution:** Approved for public release; distribution is unlimited. **Version:** 1.0 | March 2026

NOTE

All TMs in the Maven training series reference "USAREUR-AF naming conventions." This document is the authoritative source for those conventions. When in doubt, consult C2DAO or the responsible data steward.

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SECTION 1 — PROJECT NAMING

1-1. Format

```
[UNIT]-[FUNCTION]-[DESCRIPTOR]
```

Component	Description	Example
UNIT	Unit identifier (USAREUR, VCORPS, 21TSC, G2, G6, S3, etc.)	`VCORPS`
FUNCTION	Functional domain (OPDATA, LOG, PERS, INTL, PM, KM, AI)	`LOG`
DESCRIPTOR	Short, plain-English descriptor of the project's purpose	`FuelTracking`

Examples: - `VCORPS-LOG-FuelTracking` - `USAREUR-G2-ThreatAnalysis` - `21TSC-PM-ProgramDashboard` - `USAREUR-KM-LessonsLearned`

1-2. Rules

- All caps for unit and function identifiers; CamelCase or snake_case for descriptor
- No spaces — use hyphens between components
- Maximum 50 characters
- Do not include classification markings in project names
- Do not include personal names, nicknames, or humor
- Projects in the shared enterprise space require C2DAO coordination before creation

SECTION 2 — DATASET NAMING

2-1. Format

```
[tier]_[unit]_[source]_[content]
```

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Component	Description	Example
tier	Pipeline tier (raw, bronze, silver, gold)	`silver`
unit	Unit or system of record	`gcss`
source	Source system or feed	`dailyreport`
content	What the data represents	`equipment_status`

Examples: - `raw_gcss_dailyreport_equipment_status` - `silver_ipps_weekly_personnel_strength` - `gold_g2_monthly_threat_summary` - `bronze_manual_sitrep_submissions`

2-2. Rules

- All lowercase, underscores only (no hyphens, no spaces)
- Tier prefix is mandatory: `raw` → `bronze` → `silver` → `gold`
- Do not abbreviate content descriptors to the point of ambiguity
- Include unit/system of record so consumers know provenance without reading lineage
- Datasets that aggregate across units use `enterprise` as the unit component (e.g., `gold_enterprise_readiness_summary`)

SECTION 3 — OBJECT TYPE AND PROPERTY NAMING

3-1. Object Type Format

```
[Domain][Entity]
```

- PascalCase, no underscores, no hyphens
- Singular noun (not plural)
- Domain prefix for shared enterprise types; omit for unit-local types

Examples: - `LogEquipmentItem` (shared enterprise) - `PersStrengthRecord` (shared enterprise) - `MissionEvent` (unit-local — no domain prefix) - `ProgramMilestone` - `ThreatIndicator` - `LessonLearned`

3-2. Property Naming

- camelCase

- Descriptive, unambiguous
- Use standardized suffixes:

Suffix	Use	Example
`Id`	Unique identifier	`equipmentId`, `unitId`
`Date`	Calendar date (no time)	`reportingDate`, `completionDate`
`Dtg`	Date-time group	`submissionDtg`, `lastUpdatedDtg`
`Status`	Enumerated status value	`readinessStatus`, `missionStatus`
`Count`	Integer quantity	`personnelCount`, `vehicleCount`
`Rate`	Percentage or ratio (0–1 or 0–100, documented)	`missionCapableRate`
`Name`	Human-readable label	`unitName`, `programName`
`Code`	Standardized code value	`unitIdentificationCode`, `dodaac`

3-3. Rules

- Do not use generic property names: `value`, `data`, `info`, `misc` — be specific
- Date properties store ISO 8601 strings (`YYYY-MM-DD`) unless otherwise documented
- Enum values: ALL_CAPS_UNDERSCORE (e.g., `MISSION_CAPABLE`, `NOT_READY`, `UNKNOWN`)
- Boolean properties: prefix with `is` or `has` (e.g., `isMissionCapable`, `hasOverdueActions`)

SECTION 4 — LINK TYPE NAMING

4-1. Format

```
[SourceObjectType]_[relationship]_[TargetObjectType]
```

- snake_case
- Relationship verb in present tense, directional

Examples: - `LogEquipmentItem_assignedTo_Unit` - `MissionEvent_supports_Program` - `LessonLearned_appliesTo_MissionEvent` - `ProgramMilestone_hasRisk_ProgramRisk`

4-2. Rules

- Direction matters — link name should read left-to-right as a sentence fragment
- Avoid generic relationships: `relatedTo`, `linkedTo` — be specific about the relationship
- Bidirectional links are supported in Foundry; both directions should be named logically

SECTION 5 — ACTION NAMING

5-1. Format

```
[Verb][ObjectType][Qualifier]
```

- PascalCase
- Imperative verb (Submit, Update, Approve, Close, Flag, Create, Archive)

Examples: - `SubmitSitrep` - `UpdateEquipmentStatus` - `ApproveMilestone` - `FlagThreatIndicator` - `ArchiveLessonLearned` - `CreateProgramRisk`

5-2. Rules

- Verb should accurately reflect the operation (do not use `Edit` for everything — distinguish Create / Update / Delete / Submit / Approve / Archive)
- Actions that write to shared enterprise Object Types require Data Steward review before publication
- Destructive actions (Delete, Archive) require additional confirmation logic in the Action validator

SECTION 6 — WORKSHOP APPLICATION NAMING

6-1. Format

```
[Unit] [Function] [Dashboard/Form/App] v[N]
```

- Title case
- Version number (v1, v2) appended when multiple versions coexist during transition; drop version when only one exists
- "Dashboard," "Form," "Tracker," "Report" as appropriate

Examples: - `VCORPS Readiness Dashboard` - `21TSC Fuel Status Dashboard v2` - `USAREUR Program Health Report` - `G6 MSS Access Request Form` - `S3 Mission Event Tracker`

6-2. Rules

- Application title is user-facing — write for the consumer, not the builder
- Include unit identifier so users in a shared environment know the application's owner
- Applications published to shared enterprise space require C2DAO coordination
- All commander-facing dashboards must display a **data-as-of** timestamp prominently

SECTION 7 — PIPELINE AND TRANSFORM NAMING

7-1. Pipeline Builder Job Names

```
[unit]_[source]_to_[destination]_[frequency]
```

Examples: - `gcss_daily_equipment_to_bronze_daily` - `ipps_weekly_personnel_to_silver_weekly` - `enterprise_readiness_aggregation_to_gold_daily`

7-2. Code Transform Names (Python)

- snake_case function names
- Descriptive of the transformation, not the data
- Include docstring with: input dataset(s), output dataset, transformation logic summary, last-updated date

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```
@transform_df(
    Output("/USAREUR-AF/gold/enterprise_readiness_summary"),
    personnel=Input("/USAREUR-AF/silver/ipps_weekly_personnel_strength"),
    equipment=Input("/USAREUR-AF/silver/gcss_daily_equipment_status"),
)
def compute_readiness_summary(personnel, equipment):
    """
    Joins personnel strength and equipment status to produce
    enterprise readiness summary by unit.

    Inputs:
        personnel: IPPS-A weekly strength data (silver tier)
        equipment: GCSS-Army daily equipment status (silver tier)
    Output:
        Readiness summary at unit level (gold tier)
    Last updated: 2026-03
    """
```

7-3. Rules

- Do not name transforms after their author or assignment (e.g., not `smith_transform`, `s6_project_may`)
- Transforms must be idempotent — running twice on the same input produces the same output
- Incremental transforms must document the watermark column and increment logic in the docstring

SECTION 8 — BRANCH NAMING

8-1. Format

```
[type]/[unit]-[descriptor]-[YYYYMM]
```

Type	Use
`dev`	Active development branch
`review`	Submitted for Data Steward or C2DAO review
`hotfix`	Emergency production fix

Examples: - `dev/vcorps-readiness-dashboard-202603` - `review/usareur-threat-ontology-update-202603` - `hotfix/sitrep-pipeline-null-fix-202603`

8-2. Rules

- Never build directly on `main` — always use a dev branch
- Branch names must be lowercase with hyphens
- Delete branches after merge — do not accumulate stale dev branches
- Production promotion requires Data Steward sign-off documented in the review ticket
- `main` branch is the production state — treat it as operational

SECTION 9 — CODE AND REPOSITORY STANDARDS

9-1. File and Module Naming

- Python modules: `snake_case.py`
- Config files: `snake_case.yaml` or `snake_case.json`
- No spaces in any filename committed to version control

9-2. Variable and Function Naming

- Python: `snake_case` for variables and functions; `PascalCase` for classes
- TypeScript: `camelCase` for variables and functions; `PascalCase` for classes and types
- Constants: `ALL_CAPS_UNDERSCORE` in both Python and TypeScript

9-3. Commit Message Format

```
[type]: [short description] ([unit/project])

[optional body — what changed and why]
```

Types: `feat`, `fix`, `refactor`, `docs`, `test`, `chore`

Examples: - `feat: add fuel consumption forecast model (VCORPS-LOG)` - `fix: null handling in IPPS strength join (USAREUR-PERS)` - `docs: update TM-30 workshop chapter (maven-training)`

9-4. Security Rules (Non-Negotiable)

- **Never** commit credentials, connection strings, or access tokens — use environment variables or approved credential stores
- **Never** commit operational data payloads, even anonymized

- **Never** use f-strings or string concatenation to construct SQL — parameterize all queries
- All ingestion scripts must validate inputs at the boundary before processing

SECTION 10 — GOVERNANCE CHECKPOINTS

10-1. Before Creating a New Object Type in a Shared Domain

- ☐ Coordinate with C2DAO — confirm no existing Object Type covers the use case
- ☐ Define all properties, types, and enum values before creation
- ☐ Identify the responsible Data Steward
- ☐ Document expected consumers (who will query this Object Type?)
- ☐ Apply USAREUR-AF naming conventions per Section 3

10-2. Before Publishing a Workshop Application

- ☐ Data-as-of timestamp is displayed on all data-backed views
- ☐ Access controls reviewed — only authorized users can reach sensitive data
- ☐ Tested with production-representative data (not just synthetic)
- ☐ C2DAO notified if application writes back to shared enterprise Object Types via Actions
- ☐ Application title follows naming standard (Section 6)

10-3. Before Promoting a Pipeline to Production

- ☐ Transform is idempotent — tested by running twice
- ☐ Failure alerting configured (email or notification to responsible owner)
- ☐ Downstream consumers notified 24 hours before changes to existing production pipelines
- ☐ Data Steward sign-off documented
- ☐ Branch naming follows standard (Section 8)

10-4. Before Deploying Code to Production (TM-40F / TM-40C)

See the full governance checklist in the applicable TM appendix: - TM-40C Appendix A — Model Governance Checklist - TM-40F Appendix B — SWE Security Checklist

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APPENDIX A — APPROVED ABBREVIATIONS

Abbreviation	Full Term
AOR	Area of Responsibility
C2DAO	Command and Control Data and Analytics Office
CDA	Command Data Analytics
DODAAC	Department of Defense Activity Address Code
DTG	Date-Time Group
FOO	Functions on Objects
GFEBS	General Fund Enterprise Business System
GCSS	Global Combat Support System
IPPS-A	Integrated Personnel and Pay System — Army
KM	Knowledge Management / Knowledge Manager
LOGREP	Logistics Report
MSS	Maven Smart System
OPDATA	Operational Data
ORSA	Operations Research and Systems Analysis
OSDK	Ontology Software Development Kit
PERSTAT	Personnel Status
PM	Program Manager
RAG	Red-Amber-Green (status indicator)
SITREP	Situation Report
SWE	Software Engineer
USAREUR-AF	United States Army Europe and Africa
UIC	Unit Identification Code

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APPENDIX B — GOVERNANCE CHECKLIST (BEFORE PROMOTING TO PRODUCTION)

Use this checklist for any change to a production MSS project. Tailor to the type of change.

```
[ ] Naming conventions followed (this document, applicable section)
[ ] Data Steward identified and notified
[ ] C2DAO coordination completed (if required — see Section 10)
[ ] Downstream consumers notified (for pipeline changes)
[ ] Access controls reviewed
[ ] Data-as-of timestamp present (for dashboards)
[ ] Testing completed on production-representative data
[ ] Branch naming follows standard
[ ] Commit messages follow standard
[ ] No hardcoded credentials or connection strings
[ ] All SQL parameterized
[ ] Idempotency verified (for pipelines/transforms)
[ ] TM-specific governance checklist completed (TM-40C App. A / TM-40F App. B)
```

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